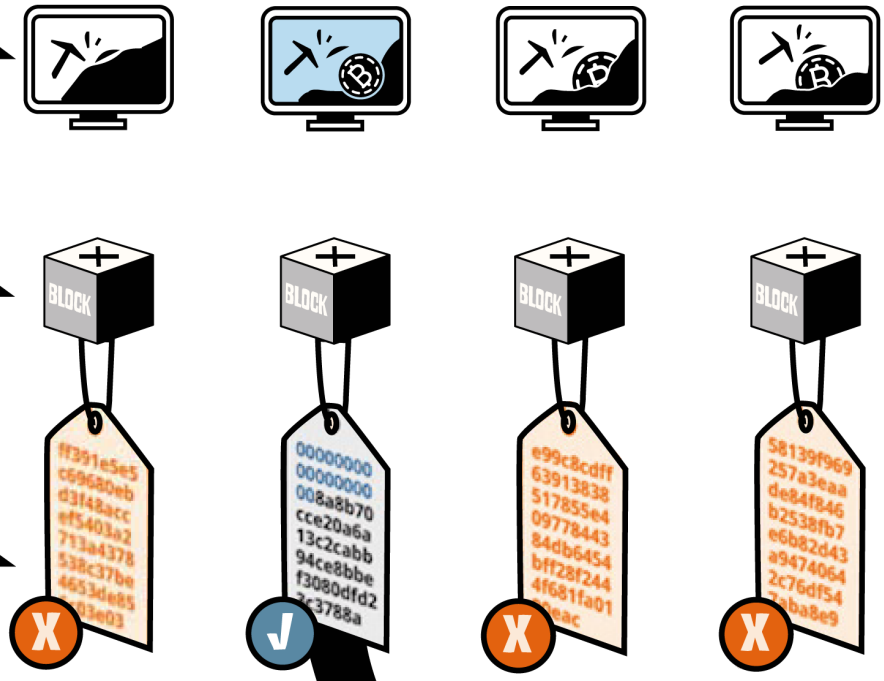
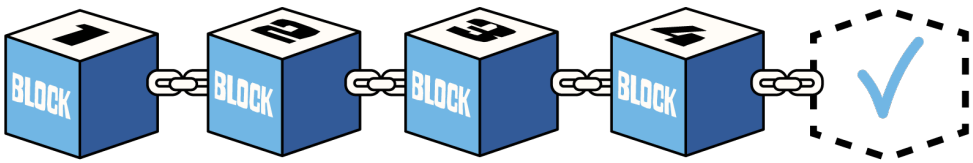
THE MINERS – The nodes working to maintain and update the ledger – gather all the latest transaction data.

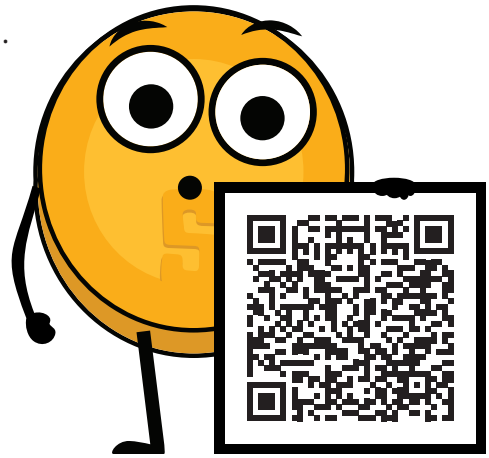
They group it all together in a **BLOCK** to add to the chain.

Then they calculate a **HASH** for that block: they run all the data through an algorithm that spits out a unique string. Then they do it again. And again. They're looking for a valid hash – one that begins with a long string of zeroes. The first miner to find one wins newly created bitcoin, and their block is added to the chain.



This hash assures no one tampers with the ledger. Changing a single number or letter anywhere in the chain will make every hash for every subsequent block **unrecognizably different**. At which other nodes will reject the compromised ledger.



6. The miner who first finds the correct hash broadcasts it to the network, and the other miners check the solution to verify that it is indeed correct.
7. If the solution is verified, the block is added to the *blockchain*.

The **mined block** is broadcasted to the network for **verification** and once it is verified by other miners and consensus is reached among the network, the block is added to the ***blockchain*** as the last block in the chain.